Language Models for Analysing Social Media Discourse: Resources and Applications

Thesis submitted in accordance with the requirements of Cardiff University for the degree of Doctor in Philosophy by

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Acronyms

AI	Artificial Intelligence
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
RoBERTa	Robustly Optimized BERT Pretraining Approach
ELMo	Embeddings from Language Models
BoW	Bag of Words
ICL	in-context learning
LDA	Latent Dirichlet Allocation
LLM	Large Language Model
LM	Language Model
MLM	Masked Language Model

PLM	Pre-trained Language Model
ML	Machine Learning
NLP	Natural Language Processing
SoTA	State of The Art
SVM	Support Vector Machine
LSTM	Long Short-Term Memory
RNN	Recurrent Neural Network
GRU	Gated Recurrent Unit
URL	Uniform Resource Locator
MTLD	Measure of Textual Lexical Diversity
GloVe	Global Vectors for Word Representation
T5	Text-to-Text Transfer Transformer
BART	Bidirectional and Auto-Regressive Transformers
Flan-T5	Fine-tuned Language Net T5
C4	Colossal Clean Crawled Corpus

1 Background

This chapter provides the theoretical and methodological foundation of the thesis, covering the evolution of Natural Language Processing (NLP) tools and their applications to social media data. It explores the challenges of processing noisy, heterogeneous text from platforms like Twitter and discusses existing work on key areas such as hate speech detection, misinformation analysis, and multilingual topic classification. By situating the thesis within the broader research landscape, this chapter demonstrates how the presented contributions advance the state of the art.

1.1 Language Models

Language Models (LMs) are fundamental to NLP, aiming to capture the statistical properties of language and predict the probability of a sequence of words. Their evolution has significantly impacted the capabilities of NLP systems, especially in handling complex human language.

1.1.1 Traditional Language Models

1.1.1.1 Early Language Models

Early LMs were often based on statistical methods, such as N-gram models, which predict the next word based on the previous N-1 words (Buck et al., 2014). These models, while foundational, suffered from limitations in capturing long-range dependencies and semantic meaning. To represent words, early systems frequently relied on static word embeddings such as Word2Vec (Mikolov et al., 2013) and Global Vectors for Word Representation (GloVe) (Pennington et al., 2014), where each word was mapped to a single, fixed vector regardless of context. While these representations improved over one-hot encodings by capturing some semantic similarity (e.g., "king" and "queen" being close in vector space), they failed to reflect how a word's meaning could shift depending on its usage in different contexts.

These limitations were overcomed, up to a degree, through the development of neural approaches capable of modeling sequential dependencies and context more effectively. Specifically, Recurrent Neural Networks (RNNs) and their variants, such as Long Short-Term Memory (LSTMs) and Gated Recurrent Units (GRUs), introduced a new class of language models capable of processing word sequences while maintaining a dynamic internal state (Yu et al., 2019; Dey and Salem, 2017). These models improved the contextual understanding of language over static embeddings by updating representations as each token in a sequence was processed. However, they still faced challenges in handling long-term dependencies and lacked efficient parallelization during training, which restricted scalability and performance

on large datasets (Sherstinsky, 2020).

1.1.1.2 Masked Language Models

Masked Language Models (MLMs) emerged with the introduction of the Transformer architecture (Subakan et al., 2021). Unlike traditional LMs that predict the next word, MLMs are trained by masking a percentage of the tokens in the input and then predicting those masked tokens based on the context provided by the unmasked tokens as seen in Figure 1.1 (Devlin et al., 2019). This bidirectional training allows them to learn richer, contextualized representations of words. These models are typically pretrained on massive corpora such as Wikipedia and the Common Crawl[1], enabling them to learn robust language representations.

For example, the embedding for the word bank in "he sat by the river bank" will differ significantly from that in "she went to the bank to deposit money," because the surrounding context informs the model about the intended sense of the word. This is in stark contrast to earlier approaches like Word2Vec or GloVe, which assign the same vector to bank regardless of its meaning in different contexts. These contextual embeddings allow MLMs to capture intricate semantic and syntactic relationships, making them highly effective across a wide range of downstream tasks.

Most contemporary MLMs are based on the BERT architecture, which is an encoder-based model leveraging bidirectional self-attention mechanisms to capture contextual information from both the left and right of each token (Devlin et al., 2019). Extensions to BERT include Robustly Optimized BERT Pretraining

[1] https://commoncrawl.org

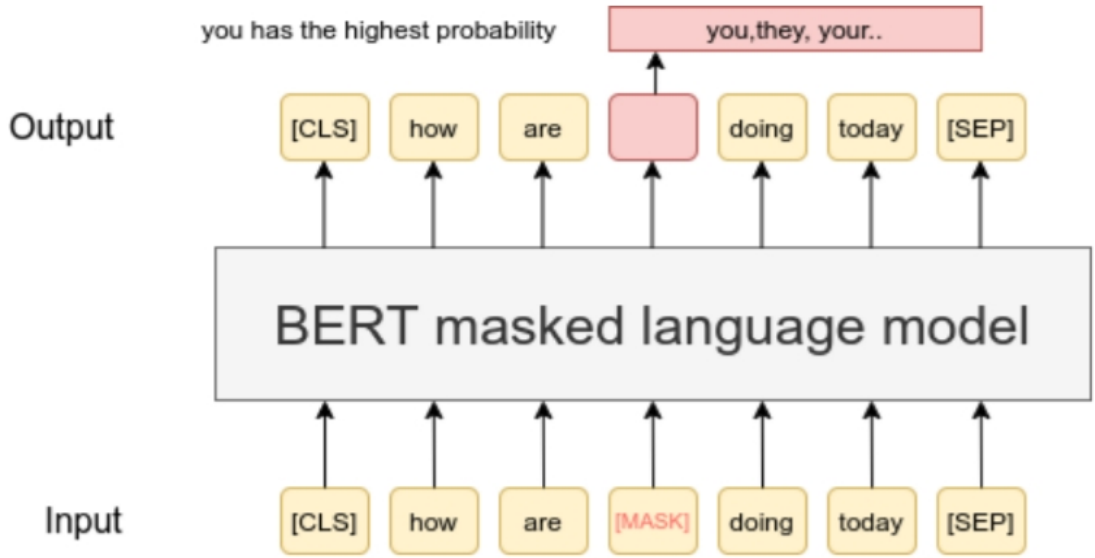

Figure 1.1: Example of MLM objective for training in Bidirectional Encoder Representations from Transformers (BERT).

Approach (RoBERTa), which optimises the pre-training process and introduces larger datasets, longer training, and XLM-R, which extends RoBERTa to multilingual settings (Liu et al., 2019; Conneau et al., 2020).

One of the main benefits of these models is that their knowledge can be utilised in various downstream tasks with minimal modifications. Also, due to their encoder architecture which allows them to understand context from both left and right, they display significantly improved performance when fine-tuned on tasks like text classification, question answering, and named entity recognition (Yang et al., 2023).MLMs have become the backbone of many state-of-the-art NLP systems, enabling significant advancements in understanding and generating human language. They are particularly effective for tasks that require deep contextual understanding, such as sentiment analysis (Alaparthi and Mishra, 2021), hate speech detection (Mozafari et al., 2019), and misinformation detection (Kaliyar

et al., 2021).

1.1.2 Encoder-Decoder Models

Encoder-decoder models represent a distinct approach in language modeling, particularly effective for sequence-to-sequence tasks. These architectures consist of two components: (i) an encoder that processes the input sequence and compresses its information into a latent representation, and (ii) a decoder that generates an output sequence based on this representation. This architecture is well-suited for tasks where the output differs in form or length from the input, such as translation, summarization, and question generation. Similar to MLMs, encoder-decoder models are also be pretrained on large corpora, while leveraging self-attention mechanisms in both the encoder and decoder stacks which allows them to learn bidirectional representations in the encoder while simultaneously training the decoder to generate sequences from noisy or incomplete inputs.

Encoder-decoder models became popular with the introduction of Text-to-Text Transfer Transformer (T5) (Raffel et al., 2020a). T5 casts NLP problems, such as summarisation, machine translation, and question asnwering, into a unified text-to-text format. It is pretrained on the Colossal Clean Crawled Corpus (C4) corpus(Raffel et al., 2020b) using a span corruption objective, where spans of text are masked and replaced with unique sentinel tokens, and the model learns to reconstruct the missing spans which enables it to develop robust representations for both understanding and generation.

An extension of T5, Fine-tuned Language Net T5 (Flan-T5) (Chung et al., 2024), further improves performance through instruction tuning (Wei et al., 2021) on a

wide variety of tasks and datasets which enhances it's zero-shot and few-shot generalization capabilities. Another notable encoder-decoder model is Bidirectional and Auto-Regressive Transformers (BART) (Lewis et al., 2019), which is trained on a denoising autoencoder objectivewhere the model learns to reconstruct original text from corrupted input. BART combines the strengths of both MLMs and autoregressive models, making it effective for tasks like summarisation and translation.

Encoder-decoder models are widely used in tasks where the input and output differ significantly in structure or length. Common applications include machine translation, text summarization, question answering, and question generation. They can also be adapted for text generation tasks, producing coherent and contextually relevant text from prompts. However, these models are generally more complex and computationally intensive than MLMs, and they not always outperform encoder-only models on tasks that do not involve sequence generation (Gasparetto et al., 2022). At the same time, they often lag behind decoder-only models, such as Large Language Models (LLMs), in open-ended text generation tasks (Saattrup Nielsen et al., 2025).

1.1.3 Large Language Models

LLMs represent the pinnacle of current NLP advancements. They are characterized by their massive scale (billions or trillions of parameters) compared to hundreds of millions usually for MLMs and similarly to them training on vast and diverse datasets encompassing text, code, as well as other modalities like images, and their ability to perform a wide range of tasks with remarkable fluency and coherence,

often in a zero-shot or few-shot learning setting ().

The majority of LLMs are also based on the Transformer architecture, but in contrast with smaller MLMs they are typically decoder-only models (), meaning they generate text by predicting the next token in a sequence without the bidirectional context used in MLMs. This allows them to excel in generative tasks, such as text completion, summarization (). Furthermore, as LLMs scale, they exhibit "emergent abilities" – capabilities that are not explicitly trained for but appear at certain model sizes, such as in-context learning or complex reasoning ().

Examples of **LLMs!**s (**LLMs!**s):

- GPT Series (Generative Pre-trained Transformer): Developed by OpenAI, particularly GPT-3 and GPT-4, are well-known for their impressive generative capabilities and wide range of applications.

- PaLM (Pathways Language Model): Google's dense decoder-only Transformer model, known for its strong performance on various language tasks.

- Llama Series: Developed by Meta, the Llama models (e.g., Llama 2, Llama 3) have gained significant traction, especially in the open-source community, due to their strong performance and accessibility for research and development.

LLMs have pushed the boundaries of what's possible in NLP, enabling more sophisticated and nuanced understanding and generation of human language.

1.2 Social media platforms

Social media platforms like Reddit and Twitter serve as rich sources of real-time, user-generated content. In general Twitter is based a form where each user can reply to another using things like posts, retweets, replies, threads, .. and is characterized by short texts (280 characters). On the other hand Reddit consists of longer, threaded discussions across topic-specific communities (subreddits), offering more contextual and hierarchical information. Despite these differences, both platforms exhibit domain-specific language and are often used to study online discourse, opinion dynamics, and harmful content.

At the same time these platforms present unique linguistic characteristics and challenges for NLPs due to their informal nature, user-generated content, character limits, prevalent use of slang, emojis, hashtags, mentions, and often noisy data.

1.2.1 Specialised Language Models for Social Media

The direct application of LMs trained on standard, clean text (like books or news articles) often underperforms on social media data. This necessitates either extensive data preprocessing or, more effectively, the development of Specialised Language Models ().

Such models are typically pre-trained or fine-tuned on large corpora of social media data which allows the models to learn the specific linguistic patterns, vocabulary, and nuances prevalent on these platforms. Specifically they are designed to handle noise and they might incorporate specific strategies to deal with typos, grammatical

errors, abbreviations, and informal language. Aware of platform-specific features Models can be designed to explicitly leverage hashtags, mentions, retweets, replies, and other structural elements of social media posts.

Examples of specialised LMs:

- BERTweet (Nguyen et al., 2020), a RoBERTa-based model pretrained on 850M English tweets, achieving strong results on sentiment analysis and hate speech detection.
- RedBERT, pretrained on Reddit data, captures stylistic and topical nuances of Reddit posts.
- TwitterRoberta
- Bernice

Models trained on domain-specific social media datasets: Many researchers pre-train or fine-tune models on large collections of tweets, Reddit posts, or other platform-specific data to improve performance on tasks relevant to those platforms, such as hate speech detection or stance detection.

The challenges of noisy, heterogeneous social media text include:

Informal Language: Slang, abbreviations (e.g., "lol", "brb"), typos, and non-standard grammar.

Contextual Ambiguity: Short posts often lack sufficient context for accurate interpretation. Sarcasm and irony are particularly difficult to detect.

Emojis and Emoticons: These visual cues carry significant emotional and semantic information.

Hashtags and Mentions: These serve as both content and metadata, indicating topics or directed communication.

Multilinguality and Code-Switching: Users often mix languages within a single post.

Data Volume and Velocity: The sheer volume and speed of social media data generation make real-time processing and analysis challenging.

Evolving Language: New slang, trends, and memes emerge rapidly, requiring models to constantly adapt.

Bias in Data: Social media data can reflect societal biases, leading to biased model outputs if not addressed carefully.

Sarcasm and Irony: Difficult to detect as the literal meaning contradicts the intended sentiment.

Implicit Sentiment: Sentiment can be expressed subtly rather than explicitly through sentiment-laden words.

Context Dependency: The same word can have different sentiment in different contexts.

Short and Noisy Text: Limited text length and informal language make it harder to extract clear sentiment signals.

Emojis and Hashtags: These often convey strong sentiment and need to be incorporated into analysis.

1.3 Tasks

NLP tasks applied to social media data extract valuable insights from the vast amount of user-generated content.

1.3.1 Sentiment Analysis

Sentiment analysis (also known as opinion mining) aims to determine the emotional tone or overall sentiment expressed in a piece of text (e.g., positive, negative, neutral). On social media, this is crucial for understanding public opinion towards brands, products, events, or political figures.

Sentiment analysis involves classifying text according to its affective polarity—positive, negative, or neutral. Social media sentiment analysis is complicated by sarcasm, slang, and domain-specific terminology.

Approaches:

Lexicon-based: Using pre-defined dictionaries of words and their associated sentiment scores.

Machine Learning (Traditional): Training classifiers (e.g., Naive Bayes, SVM) on labeled data with features like N-grams, TF-IDF.

Deep Learning (Modern): Employing RNNs, LSTMs, and especially Transformers (e.g., BERT-based models fine-tuned for sentiment analysis) that can capture complex contextual patterns.

In this thesis we utilise heavily sentiment analysis to extract insights from social media posts mainly in Chapters **??**, **??** ,and **??**.

1.3.2 Emotion Detection

While sentiment analysis typically classifies text into broad categories (positive/negative/neutral), emotion detection aims for a more granular understanding of human emotions, identifying specific emotions like anger, joy, sadness, fear, surprise, disgust, etc. This is particularly relevant for understanding emotional responses to events or content on social media.

Beyond sentiment polarity, emotion detection aims to identify specific emotional states (e.g., anger, joy, sadness). Annotated corpora such as GoEmotions (**?**) have enabled supervised learning in this space. However, emotional nuance and subjectivity make this task challenging, especially in multilingual settings.

Challenges on Social Media:

Similar to sentiment analysis, but with the added complexity of distinguishing between finer emotional nuances.

Lack of large, diverse datasets labeled with fine-grained emotions.

Cultural and linguistic variations in emotion expression.

Approaches:

Often relies on annotated datasets (e.g., GoEmotions, EmoContext) and deep learning models, particularly those based on Transformer architectures, which are well-suited for capturing the subtle linguistic cues associated with different emotions.

Lexicon-based methods for specific emotion words or phrases.

Chapters **??**, and **??**.

1.3.3 Topic Classification

Topic classification (or text categorization) assigns one or more predefined categories or topics to a given text. On social media, this helps in organizing vast amounts of information, identifying trending discussions, or filtering content based on specific themes.

Topic classification assigns thematic labels to texts. On social media, this task supports content moderation, trend analysis, and community profiling. Traditional approaches rely on supervised classification, while unsupervised methods like topic modeling (e.g., LDA) remain useful in low-resource contexts.

Challenges on Social Media:

Short Text: Tweets or short posts often don't provide enough information for accurate topic classification using traditional methods.

Dynamic Topics: Topics on social media can emerge and disappear quickly.

Multilingual Content: Discussing the same topic across different languages.

Approaches:

Traditional Machine Learning: Classifiers trained on features like bag-of-words or TF-IDF.

Topic Modeling: Unsupervised methods like Latent Dirichlet Allocation (LDA) or Non-negative Matrix Factorization (NMF) can discover underlying topics in a corpus. However, these often struggle with very short texts.

Deep Learning: Transformer-based models fine-tuned on labeled social media data for specific topic categories have shown superior performance.

Multilingual Topic Classification: Requires either language identification and separate models for each language, or multilingual embeddings and models capable of handling code-switched text.

Chapters ??, ??,??,

1.3.4 Hate Speech Detection

Hate speech detection is a critical application of NLP on social media, aiming to identify content that expresses hatred or encourages violence towards individuals or groups based on attributes like race, ethnicity, religion, gender, sexual orientation, etc. This is a highly sensitive and challenging task due to its subjective nature and

the evolving tactics of hate speech perpetrators.

Hate speech detection involves identifying offensive, threatening, or discriminatory language. Prior work uses keyword matching, lexical features, and deep learning-based classifiers. Recent benchmarks such as HatEval and datasets like HASOC provide resources for multilingual and cross-platform evaluation.

Challenges on Social Media:

Ambiguity and Context: Distinguishing genuine hate speech from offensive language, sarcasm, or legitimate criticism.

Evolving Language: Hate speech uses slang, coded language (dog whistles), and novel expressions to evade detection.

Imbalance in Data: Hate speech is typically a minority class, leading to skewed datasets and model bias towards the majority (non-hate speech) class.

Subjectivity of Definitions: What constitutes "hate speech" can vary across jurisdictions and cultural contexts.

Multilingualism: Hate speech manifests differently in various languages and cultures.

Approaches:

Rule-based Systems: Using keywords and pattern matching, but easily circumvented.

Machine Learning: Training classifiers on various linguistic features (n-grams,

sentiment lexicons, parts-of-speech tags).

Deep Learning: Particularly effective, using CNNs, LSTMs, and Transformer models (e.g., BERT, RoBERTa fine-tuned for hate speech detection). These models can capture complex semantic and syntactic patterns.

Multi-modal Approaches: Combining text with image or video analysis (though primarily text-focused in most research).

Weak Supervision/Semi-supervised Learning: Leveraging large amounts of unlabeled data, or noisy labels (e.g., from user reports) to augment smaller, carefully annotated datasets.

Ethical Considerations: Crucial for responsible development and deployment, including mitigating bias, ensuring transparency, and considering potential for censorship or chilling effects on free speech.

1.3.5 Misinformation & Polarisation

The spread of misinformation (unintentional falsehoods) and disinformation (intentional falsehoods) and its role in fostering polarisation on social media platforms pose significant societal challenges. NLP plays a crucial role in analyzing and combating these phenomena.

The spread of misinformation on social media contributes to public harm, especially in contexts like health communication and political discourse. Identifying false or misleading content is complicated by subtle rhetorical strategies and the rapid evolution of narratives.

Polarisation further exacerbates misinformation spread, as users form echo chambers and engage in selective exposure. Language models can help identify linguistic signals of polarisation and frame detection. However, challenges remain in detecting emerging misinformation, handling multimodal content, and mitigating algorithmic biases.

Misinformation and Disinformation Analysis:

Detection: Identifying false or misleading content. This often involves:

Fact-checking: Verifying claims against reliable sources (challenging to automate fully).

Stance Detection: Determining if a text agrees, disagrees, discusses, or is unrelated to a given claim.

Propaganda Detection: Identifying rhetorical techniques used to manipulate public opinion.

Source Credibility: Assessing the reliability of the source of information.

Linguistic Features: Analyzing linguistic patterns (e.g., sensationalist language, emotional intensity, specific vocabulary) often associated with misinformation.

Propagation Analysis: Understanding how misinformation spreads across networks, often leveraging network science alongside NLP to analyze user interactions, sharing patterns, and community structures.

Debunking and Intervention: Developing NLP-powered systems to identify and flag misinformation, or to generate counter-narratives and provide accurate information.

Polarisation Analysis:

Echo Chambers and Filter Bubbles: NLP can identify communities with shared beliefs and analyze how information flows within and between these groups.

Opinion Dynamics: Tracking how opinions evolve over time and how they are influenced by social interactions and exposure to different narratives.

Stance and Ideology Detection: Classifying users or content based on their political or social leanings.

Linguistic Markers of Polarisation: Analyzing specific language choices (e.g., divisive rhetoric, "us vs. them" framing, emotional language) that indicate or contribute to polarization.

Disagreement and Deliberation: Studying how online discussions devolve into unproductive arguments versus constructive dialogue.

Challenges in Misinformation & Polarisation:

"Truth" Definition: The ground truth for misinformation can be ambiguous or rapidly change.

Evolving Tactics: Misinformation campaigns constantly adapt their language and dissemination strategies.

Subtlety: Misinformation can be subtle, relying on implications or omissions rather than outright falsehoods.

Emotional Resonance: Misinformation often leverages strong emotions, making it

highly shareable.

Cross-platform Coordination: Campaigns often span multiple social media platforms.

Attribution: Identifying the source and intent behind misinformation is complex.

Ethical Implications: The risk of suppressing legitimate dissent or creating "thought police" when developing automated detection systems.

Approaches:

Hybrid Models: Combining NLP techniques with network analysis, computer vision (for multimodal misinformation), and knowledge graphs.

Adversarial Training: Developing models robust to subtle manipulations designed to evade detection.

Explainable AI (XAI): Making models more transparent to understand why they classify certain content as misinformation or polarizing.

Large Language Models: Their ability to reason, summarize, and understand complex contexts is being explored for tasks like fact-checking assistance and generating counter-narratives. However, LLMs can also generate convincing misinformation, highlighting the dual-use nature of the technology.

work based on misinformation is done in chapter **??** while Chapters **??** **??** focuses on polarisation in social discourse.

2 Conclusions and Future Work

2.1 Summary of Thesis

2.2 Summary of the Chapters

2.3 Future Work

Bibliography

Shivaji Alaparthi and Manit Mishra. 2021. Bert: A sentiment analysis odyssey. *Journal of Marketing Analytics*, 9(2):118–126.

Christian Buck, Kenneth Heafield, and Bas Van Ooyen. 2014. N-gram counts and language models from the common crawl. In *Proceedings of the Language Resources and Evaluation Conference 2014*, pages 3579–3584.

Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Yunxuan Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, et al. 2024. Scaling instruction-finetuned language models. *Journal of Machine Learning Research*, 25(70):1–53.

Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451, Online. Association for Computational Linguistics.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.

Rahul Dey and Fathi M Salem. 2017. Gate-variants of gated recurrent unit (gru) neural networks. In *2017 IEEE 60th international midwest symposium on circuits and systems (MWSCAS)*, pages 1597–1600. IEEE.

Andrea Gasparetto, Matteo Marcuzzo, Alessandro Zangari, and Andrea Albarelli. 2022. A survey on text classification algorithms: From text to predictions. *Information*, 13(2):83.

Rohit Kumar Kaliyar, Anurag Goswami, and Pratik Narang. 2021. Fakebert: Fake news detection in social media with a bert-based deep learning approach. *Multimedia tools and applications*, 80(8):11765–11788.

Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Ves Stoyanov, and Luke Zettlemoyer. 2019. Bart: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. *arXiv preprint arXiv:1910.13461*.

Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*.

Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. 2013. Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*.

Marzieh Mozafari, Reza Farahbakhsh, and Noel Crespi. 2019. A bert-based transfer learning approach for hate speech detection in online social media. In *International conference on complex networks and their applications*, pages 928–940. Springer.

Dat Quoc Nguyen, Thanh Vu, and Anh Tuan Nguyen. 2020. BERTweet: A pre-trained language model for English tweets. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 9–14, Online. Association for Computational Linguistics.

Jeffrey Pennington, Richard Socher, and Christopher Manning. 2014. GloVe: Global vectors for word representation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1532–1543, Doha, Qatar. Association for Computational Linguistics.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. 2020a. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine learning research*, 21(140):1–67.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2020b. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67.

Dan Saattrup Nielsen, Kenneth Enevoldsen, and Peter Schneider-Kamp. 2025. Encoder vs decoder: Comparative analysis of encoder and decoder language models on multilingual NLU tasks. In *Proceedings of the Joint 25th Nordic Conference on Computational Linguistics and 11th Baltic Conference on Human Language Technologies (NoDaLiDa/Baltic-HLT 2025)*, pages 561–572, Tallinn, Estonia. University of Tartu Library.

Alex Sherstinsky. 2020. Fundamentals of recurrent neural network (rnn) and long short-term memory (lstm) network. *Physica D: Nonlinear Phenomena*, 404:132306.

Cem Subakan, Mirco Ravanelli, Samuele Cornell, Mirko Bronzi, and Jianyuan Zhong. 2021. Attention is all you need in speech separation. In *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 21–25. IEEE.

Jason Wei, Maarten Bosma, Vincent Y Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M Dai, and Quoc V Le. 2021. Finetuned language models are zero-shot learners. *arXiv preprint arXiv:2109.01652*.

Linyi Yang, Shuibai Zhang, Libo Qin, Yafu Li, Yidong Wang, Hanmeng Liu, Jindong Wang, Xing Xie, and Yue Zhang. 2023. GLUE-X: Evaluating natural language understanding models from an out-of-distribution generalization perspective. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 12731–12750, Toronto, Canada. Association for Computational Linguistics.

Yong Yu, Xiaosheng Si, Changhua Hu, and Jianxun Zhang. 2019. A review of recurrent neural networks: Lstm cells and network architectures. *Neural computation*, 31(7):1235–1270.